$$\Delta = \frac{\Delta C}{\Delta S} = \frac{\delta C}{\delta S} = 0.5$$ ① ② # of shares.

0.5

long Δ share. ↑1 in S. $1 \times 0.5 = 0.5$

short 1 call ↓ $0.5 \times 1 = 0.5$.

$S = \$100$. $C = \$10$. $\Delta = \frac{\delta C}{\delta S} = 0.6$.

$\delta S = \$1$. $\delta C = \$0.6$

$\delta C = \Delta \cdot \delta S = 0.6 \times 1 = 0.6$

Short 2000 Call ↓

Long ? S ↑

1,200

Loss = $2000 \times 0.6 = 1200$

$\# = 0.6 \times 2000 = 1200$

Gain = $1,200 \times \$1 = \$1200$

$\Delta S = \$2$.

$\delta C = \Delta \cdot \delta S = 0.6 \times 2 = 1.2$

Short 2000 call

Long 1200 S

Loss = $2000 \times 1.2 = 2,400$

Gain = $1,200 \times 2 = 2,400$

Long call
↑S → ↑C

Short call

Long put ↓
↑S → ↓P

Short Put
↑S → ↑P

Short S
↓
Long S

Long S

Short S

	Δ	Position in S
Long call	$0 < N(d_1) < 1$	Short $N(d_1)$ S
Short call	$-N(d_1) < 0$	Long $N(d_1)$ S
Long put	$N(d_1) - 1 < 0$	Long $1 - N(d_1)$ S
Short put	$1 - N(d_1)$	Short $1 - N(d_1)$ S

Short $-N(d_1)$ S

Short $N(d_1) - 1$ S

div- paying S.

Long call. $e^{-qT} \cdot N(d_1)$

Short call $-e^{-qT} N(d_1)$

Long put $e^{-qT} \cdot (N(d_1) - 1)$

Short put $e^{-qT} (1 - N(d_1))$

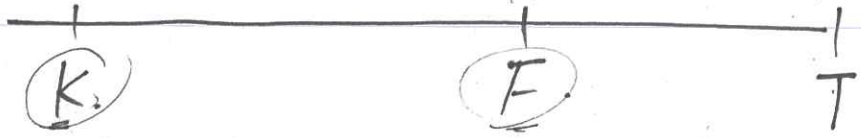
Options on Futures.

e^{-qT} div- paying.

e^{-rfT} F.X

e^{-rT} Futures options.

long. to



$$f = (F - K) \cdot e^{-rT}$$ long forward.

$$F = S \cdot e^{rT}$$ $$F = S \cdot e^{(r-q)\cdot T}$$

$$f = (S \cdot e^{rT} - K) \cdot e^{-rT} = S - K \cdot e^{-rT}$$

$$\Delta = \frac{\delta f}{\delta S} = 1$$

---

non-div S.

	Δ	
Long forward	+1	Short 1 S.
Short forward	-1	Long 1 S.

$$f = (S \cdot e^{(r-q)\cdot T} - K) \cdot e^{-rT} = S \cdot e^{-qT} - K \cdot e^{-rT}$$

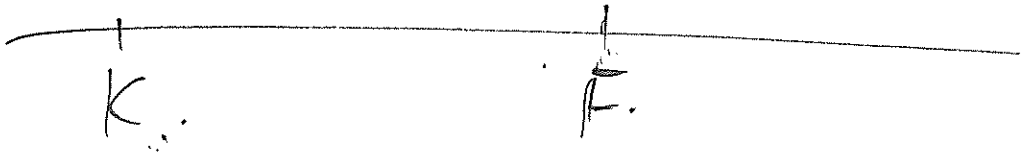
$$\Delta = e^{-qT}$$

---

div-paying S.

Long forward	e^{-qT}	Short e^{-qT} S
Short forward.	$-e^{-qT}$	Long e^{-qT} S.

long.



$f = F - K$. long futures.

$= S \cdot e^{rT} - K$.

$\Delta_{futures} = e^{rT}$ long F.

$f = S \cdot e^{(r-q)T} - K$.

$\Delta_{futures} = e^{(r-q)\cdot T}$ long F

Short put. Δ-neutral. short 978k pounds.

↑ replace.

Δ-neutral. ~~long~~ Short? Futures. 991,540

S↑ short ~~long~~ F ↓ Short Put ↑.

$$\Delta_{futures} = e^{(r - r_f)\cdot T}$$

For 1 ~~unit of~~ F, Δ units of S.

$$\Delta = e^{(0.06 - 0.075) \times \frac{11}{12}} = 0.9863.$$

$$\frac{1}{X} = \frac{0.9863}{978k}$$

1 Future.	0.9863 pounds.
X	978k pounds.

$X = 978k / 0.9863. = .991,540$

Question 1

A put option on the ASX200 has an exercise price of 5,000 and maturity of 1 year. The risk-free interest rate is 7% p.a. The dividend yield on the index is 3% p.a. The volatility is 20% p.a. and the current level of the index is 5,000. A financial institution has a short position in the option.

a) Calculate the delta of a short position in this put option.

b) How can the position be made delta neutral?

c) Suppose that one hour later the index has increased to 5,150, how can delta neutrality be preserved?

$K = 5000 \quad T = 1 \quad r = 7\% \quad q = 3\% \quad \sigma = 20\% \quad S = 5000$

a) Δ for short put.

$$\Delta = (1 - N(d_1)) \cdot e^{-qT}$$

$$d_1 = \frac{\cdot \quad \cdot (\cdot\cdot - q\cdot\cdot)}{\cdot \quad \cdot\cdot}$$

$\Delta = 0.371$

b) For each short put (on a single of unit of S.), take short position in 0.371 unit of index.

c) $S_1 = 5150$.

one month later. $T = \frac{11}{12}$

$\Delta = 0.317$.

Short $0.317 \cdot S$.

Long $0.371 - 0.317 = 0.054$.

Question 2

Suppose a financial institution has the following three positions in options on a stock:

1) A long position in 100,000 call options with strike price $55 and an expiration date in 3 months. The delta of this long call is 0.533.

2) A short position in 200,000 call options with strike price $56 and an expiration date in 5 months. The delta of this short call is -0.468.

3) A short position in 50,000 put options with strike price $56 and an expiration date in 2 months. The delta of this short put is 0.508.

How can you make your portfolio delta neutral?

$0.533 \times 100k + (-0.468 \times 200k) + 0.508 \times 50{,}000$

$= -14{,}900$

$\uparrow S$

Long position in S. 14,900 unit.